

Characterizing the Infralow Fluctuation of Sigma Power during N2 Sleep in Healthy Aging and Mild Cognitive Impairment

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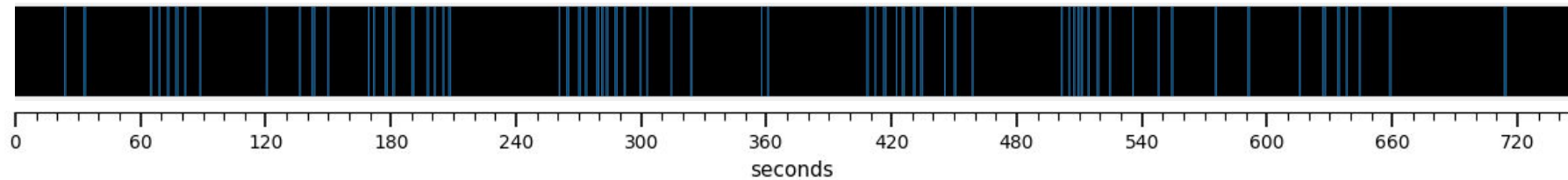
Introduction: Intraslow Fluctuation of Sigma Power (ISFS)

The Phenomenon

- *Definition:* NREM sleep is organized by a highly conserved **~0.02 Hz oscillation** in EEG sigma power (10-16 Hz).
- *Microstructure:* This rhythm defines distinct alternating substates of **"Stable" sleep** (high sensory gating) and **"Fragile" sleep** (susceptibility to micro-arousals) (*Lecci et al., 2017; Lázár et al., 2019*).

Mechanisms & Goals

- *Noradrenergic Driver:* These fluctuations are paced by the **Locus Coeruleus (LC)**; infraslow rhythms in noradrenergic activity modulate the thalamocortical loops that generate spindles (*Osorio-Forero et al., 2021*).



Visualizing the ISFS Rhythm: 12.5 minute N2 sleep trace demonstrating the natural clustering of spindle events (blue bars) on an infraslow timescale.

Current Study: ISFS in Healthy Aging and MCI

- *The Framework:* Implementation of the ISFS analysis pipeline established by **Dimitriades et al. (2024)** to extract and characterize specific temporal parameters of the fluctuation during **N2 sleep**.
- *Project Scope:* Extends the original developmental framework to a clinical aging context by evaluating ISFS characteristics across **Young Adult Controls, Healthy Elderly, and MCI** groups within international datasets from Tel Aviv and Sydney.

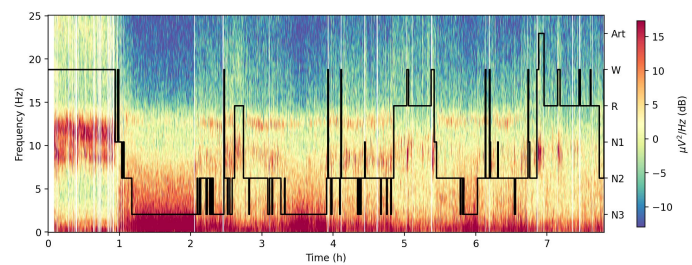
Cohort Demographic and Exclusion Reasons

Variable	Young	Elderly	MCI
N (analyzed)	34	38	31
Age (years)	27.7 ± 5.1	66.5 ± 9.9	67.8 ± 9.0
MoCA (where available)	—	26.9 ± 2.5 (n=18)	21.0 ± 4.1 (n=14)
% bad channels	6.3 ± 3.5	1.4 ± 1.8	3.4 ± 4.4
% bad epochs (in N2)	4.6 ± 5.2	3.4 ± 4.0	1.8 ± 2.0
Exclusions			
Total excluded	11	5	10
Not enough N2	5	4	4
Low ISFS detection rate	4	0	0
Too many bad channels	1	0	4
Too many bad epochs	1	1	2

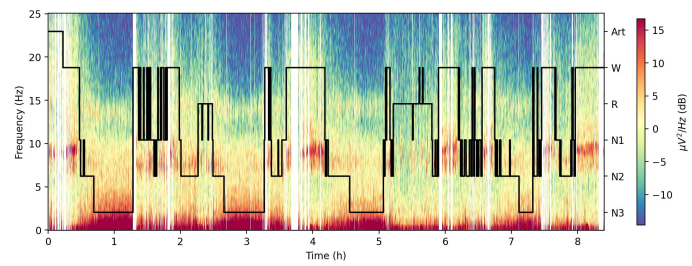
Values are mean ± SD. MoCA (n) = subjects with a MoCA score. % bad epochs computed within N2 sleep.

Sleep Architecture Across Study Groups

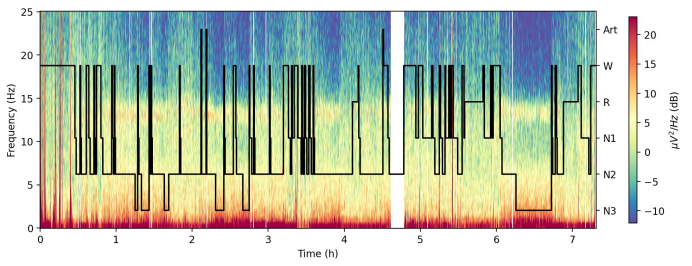
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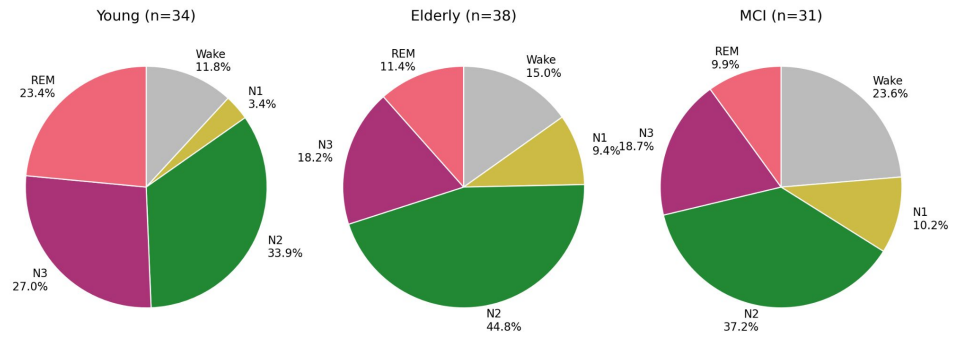
E



MCI



Sleep stage distribution (% of total recording, group mean)



Stage	Omnibus p	Young vs Elderly	Young vs MCI	Elderly vs MCI
Wake	<.001	* (.034)	*** (<.001)	* (.034)
N1	<.001	*** (<.001)	*** (<.001)	ns (.413)
N2	<.001	*** (<.001)	ns (.299)	* (.018)
N3	<.001	*** (<.001)	** (.002)	ns (.976)
REM	<.001	*** (<.001)	*** (<.001)	ns (.688)

*** p<.001 ** p<.01 * p<.05 ns = not significant | post-hoc: Tukey HSD if all-normal, else Dunn (Holm)

Aging compresses deep and REM sleep; MCI fragments it further, yet N2 (where ISFS lives) remains plentiful in all three groups, ruling out simple "not enough N2" as the driver of any downstream ISFS difference.

N2 Bout Selection & Characteristics

- *Artifact Cleaning:* Bad epochs are removed from scored N2 segments.
- *Duration Threshold:* Only clean N2 bouts **longer than 300s** are included, as the ISFS operates on a 10-100s timescale, a minimum window of 300s is required to capture at least 3–5 full cycles of the infraslow oscillation.

N2 Bout Properties Group Comparison

Variable	Young	Elderly	MCI	Omnibus p	Y vs E	Y vs MCI	E vs MCI
Bout count	7.59 ± 3.92	11.13 ± 4.13	10.32 ± 4.98	.002	.002	.034	.722
Mean Bout Length (s)	635.4 ± 173.2	564.9 ± 106.8	505.2 ± 79.4	<.001	.053	<.001	.130
Total Duration of Bouts (min)	80.8 ± 45.4	107.8 ± 49.4	86.7 ± 42.4	.036	.039	.863	.147
Mean Relative Location (%)	55.7 ± 13.3	54.2 ± 6.7	53.7 ± 10.2	.126	—	—	—
Proportion of all N2 (%)	52.3 ± 16.0	53.0 ± 17.0	51.1 ± 15.4	.889	—	—	—

Bout = clean NREM2 segment ≥ 300 s, split around BAD epochs. Values are mean ± SD.
Omnibus: ANOVA if all groups normal, else Kruskal-Wallis. Post-hoc: Tukey HSD or Dunn (Holm). Bold = p<.05.

Observations

- *Morphological Changes:* Aging and MCI exhibit a "fragmented" N2 profile, characterized by more frequent but shorter stable bouts compared to Young Adults.
- *Balanced Representation:* Despite differences in bout morphology, the total **Proportion of N2** sampled remains consistent across groups.
- *Future Directions:* To ensure these duration differences do not bias our spectral results, we are considering a secondary sensitivity analysis using an equalized/reduced bout count across cohorts.

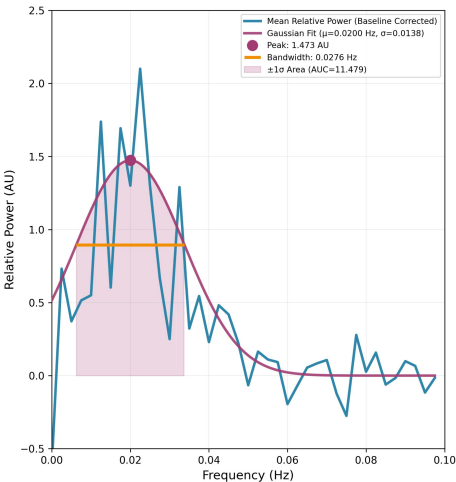
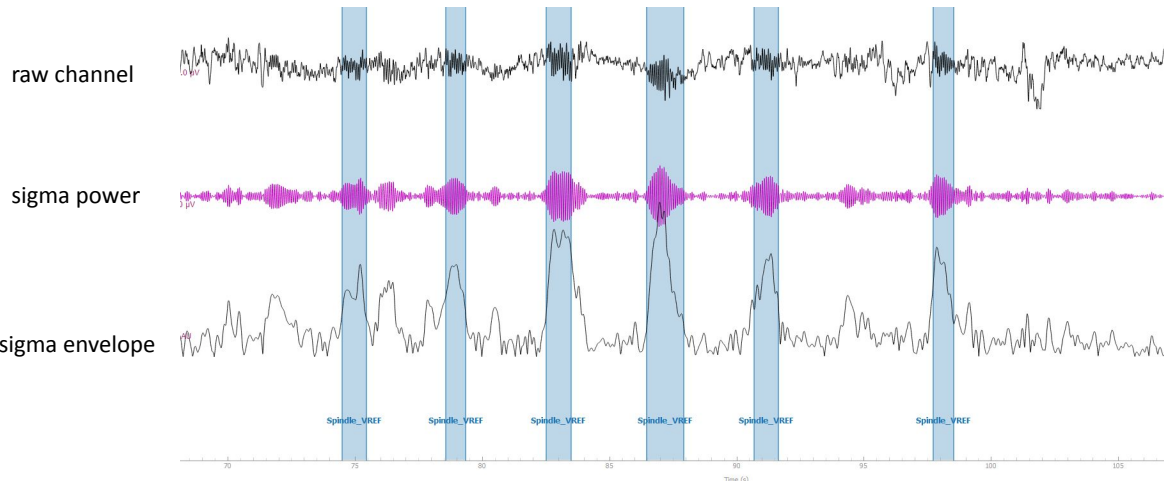
Quantifying the Rhythm – Spectral Analysis

Feature Extraction

- For each bout, calculate **Sigma Power** (13-16 Hz), its **envelope**, and **FFT** (fast Fourier transform).
- A **Gaussian** is fitted to each channel based on the average of all its bouts.

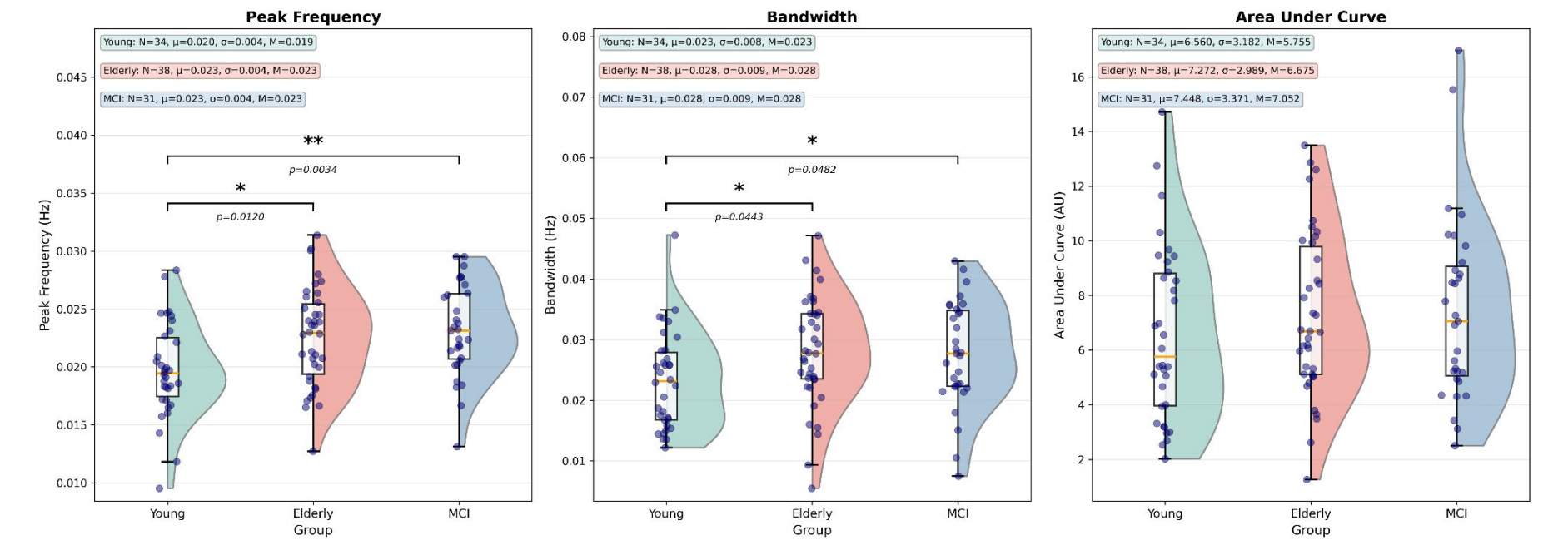
When the fit is successful, it provides three *metrics for analysis*:

- a. **Peak Frequency**: The dominant rate of the rhythm → purple dot at the peak of the gaussian.
- b. **Bandwidth (BW)**: The frequency spread; reflects rhythmic consistency → orange horizontal line spanning the curve's width.
- c. **Area Under the Curve (AUC)**: Represents the ISFS Strength → shaded area beneath the gaussian curve.



Results - ISFS Parameters Comparison

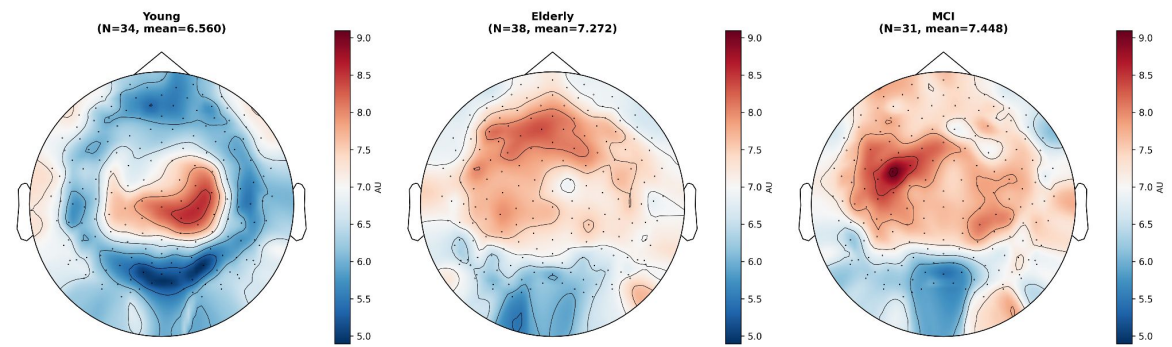
Whole-scalp ISFS parameters across groups. Each dot = one subject's mean across successfully fitted electrodes; box = IQR with median; half-violin = kernel density. Young N=34, Elderly N=38, MCI N=31.



Aging shifts the temporal properties of the ISFS rhythm, peak frequency and bandwidth, without changing its overall strength (AUC). Both elderly and MCI differ from young adults; the two older groups do not differ from each other.

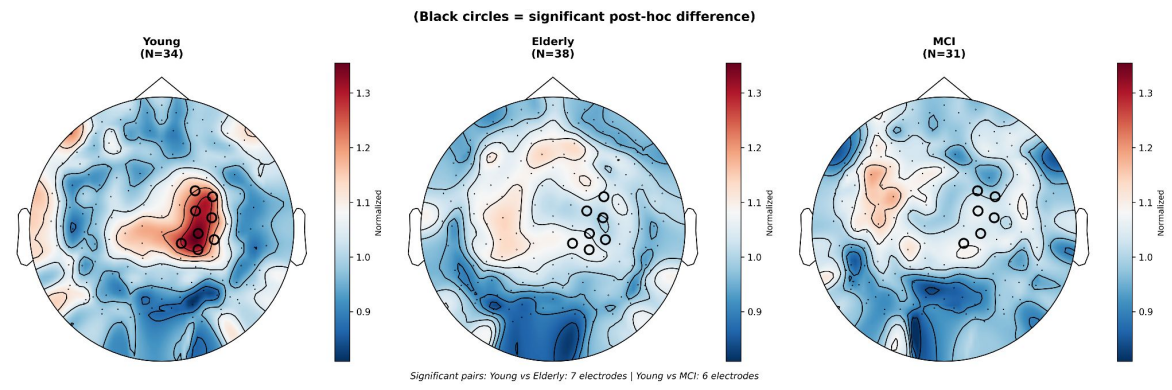
Results - AUC Topographies Across Groups

Raw values



Young adults concentrate ISFS strength over central-parietal electrodes; in aging and MCI this hotspot **flattens and spreads**.

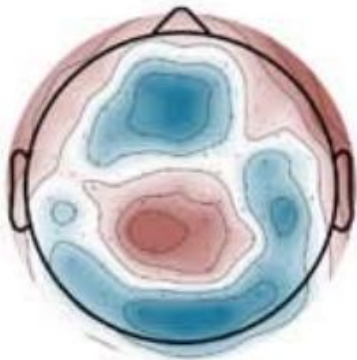
Normalized by-subject values



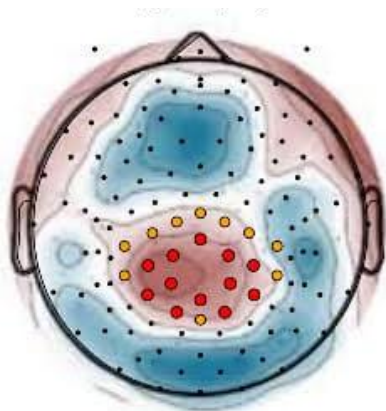
Cluster-based permutation test results in a single significant cluster (central-parietal) that captures the effect, driven by Young < Elderly and Young < MCI. Elderly and MCI do not differ.

AUC Region of Interest

Defined by **Dimitriades et al. (2024)** results for young adults' AUC:



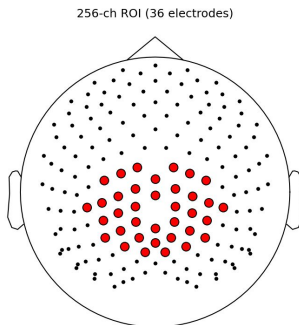
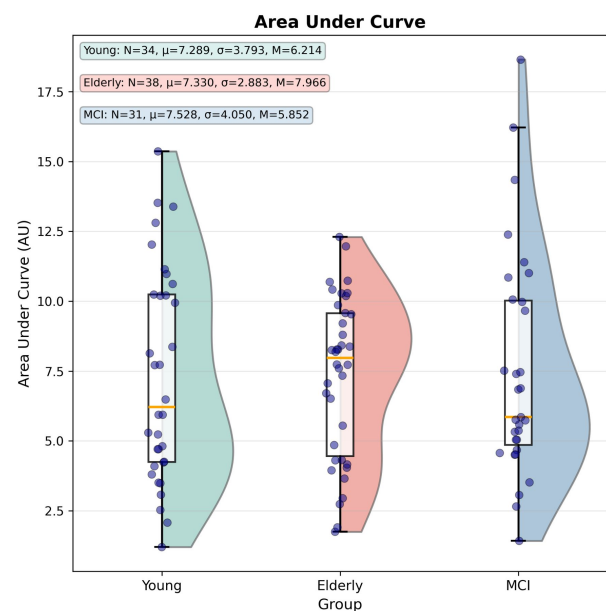
Chose the corresponding electrodes from the 128-channels system and converted them to our 256-channels system



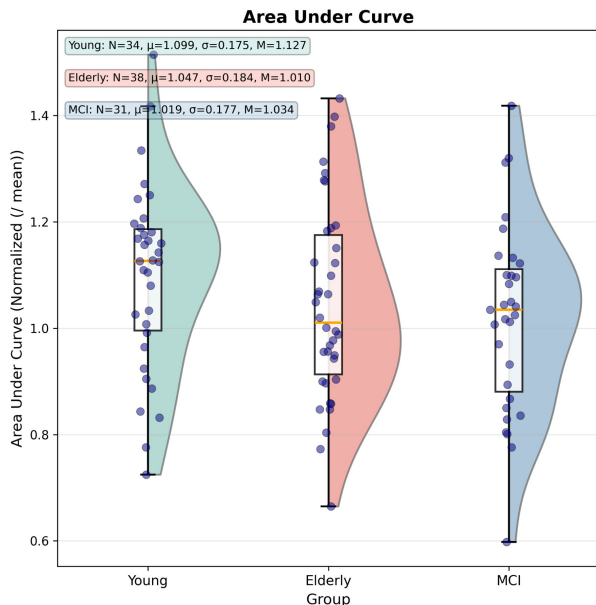
Results - AUC Region of Interest

Each dot is a subject - an average of the **ROI** successful electrodes

Raw values

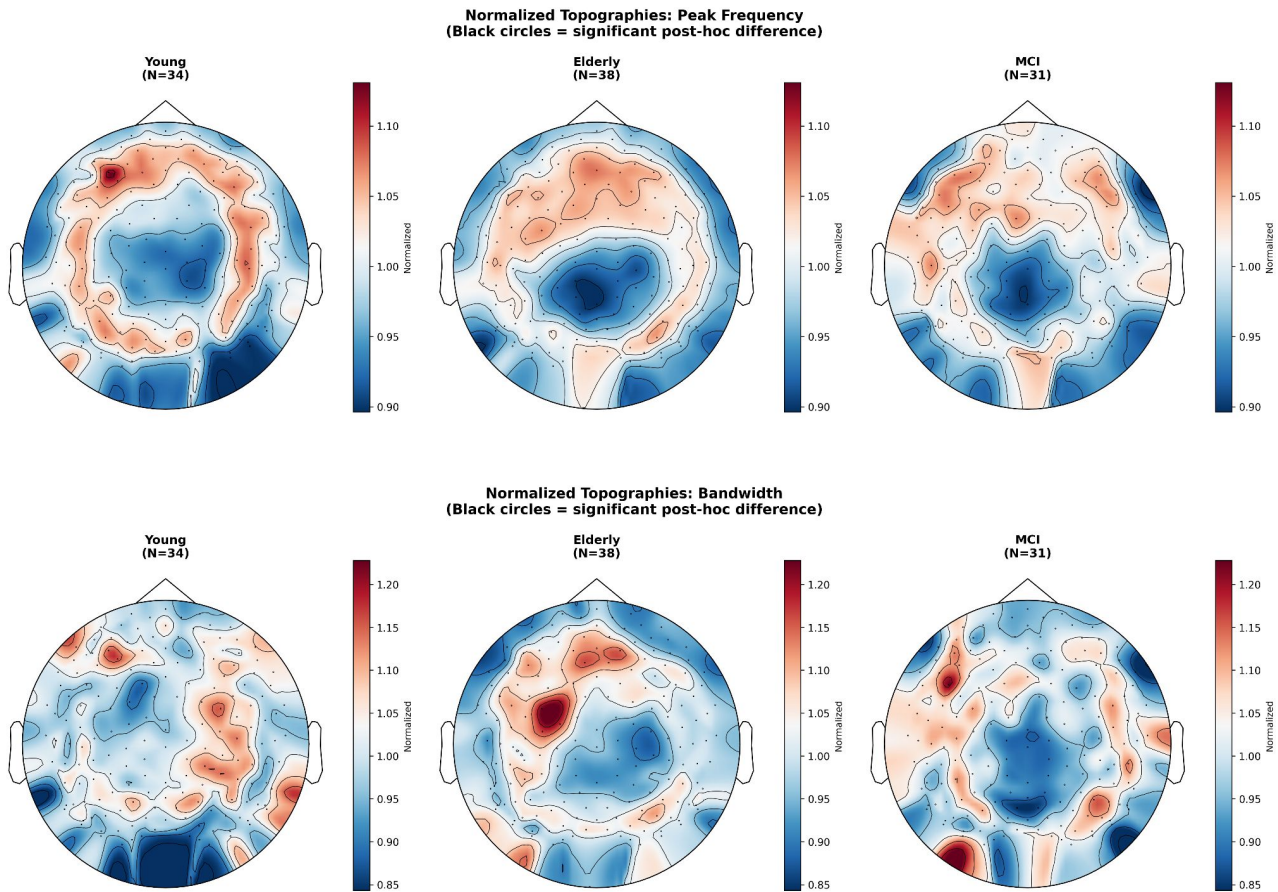


Normalized by-subject values



In the raw view the three groups are indistinguishable. The normalized view shows the expected ordering - Young preferentially concentrate AUC in this ROI ($\mu \approx 1.10$), Elderly less so ($\mu \approx 1.05$), MCI least ($\mu \approx 1.02$), but the 3-group omnibus does not reach significance. The cluster-based test on the previous slide picks up this effect; averaging across the full a-priori ROI dilutes it.

Results - Peak Frequency & Bandwidth Normalized Topographies



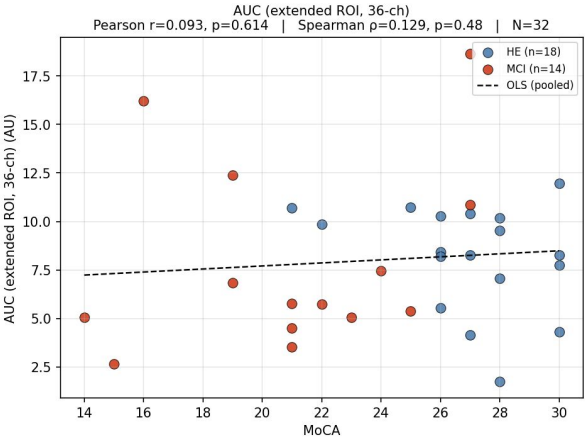
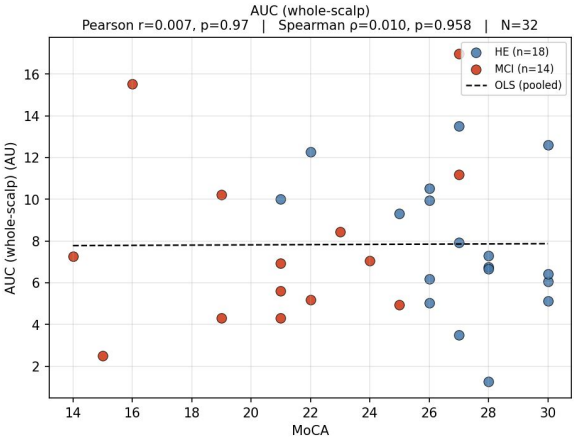
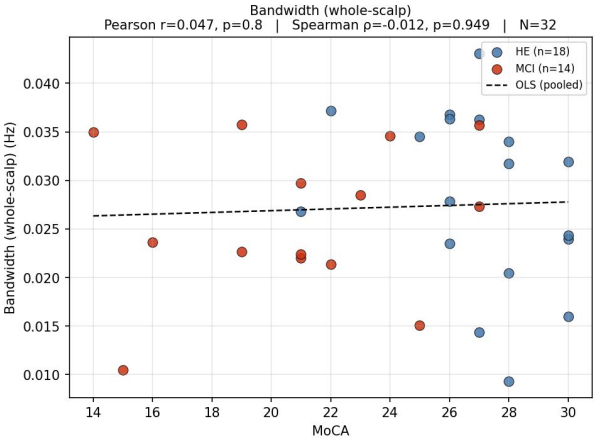
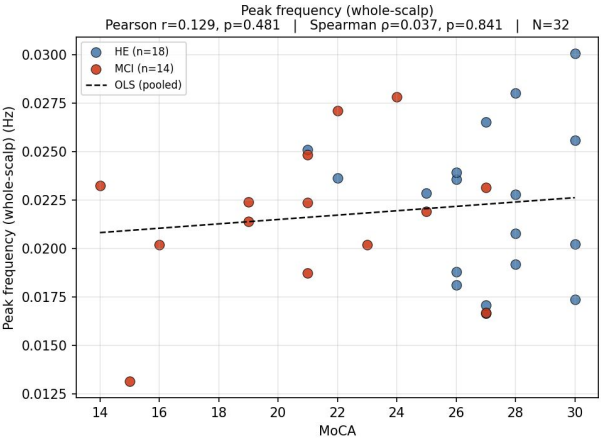
Peak frequency shows a frontal-anterior emphasis in young adults that flattens with age.

Bandwidth shows no consistent spatial pattern across the three groups;

Both metrics differ at the whole-scalp level (slide 7) but show **no cluster-significant** spatial effect. The effects are diffuse rather than regionally localized.

Results - ISFS Parameters & MoCA Correlation

ISFS scalars vs MoCA (HE+MCI pooled)



No ISFS scalar (peak frequency, bandwidth, whole-scalp AUC, or ROI AUC) correlates with MoCA in the pooled Elderly+MCI sample ($N=32$; all $|r| \leq 0.13$, all $p \geq 0.48$; Pearson and Spearman).

Conclusion

The ISFS rhythm of N2 sleep is altered, not abolished, in healthy aging. We did not find any differences between MCI and age-matched elderly within our sample.

- *Temporal alterations* - The infralow rhythm becomes faster (peak frequency: $Y < E = MCI$, $p = 0.002$) and broader (bandwidth: $Y < E = MCI$, $p = 0.025$) with age, consistent with a less precise thalamocortical clock.
- *Spatial (not global) loss of ISFS strength* - Whole-scalp AUC does not differ between groups, but cluster-permutation analysis reveals a central-parietal cluster where AUC is reduced in Elderly and MCI versus Young ($p = 0.014$, 9 electrodes). The characteristic young-adult hotspot (Dimitriadis 2024) flattens with age.
- *MCI $\approx$ Elderly* - Across every test, Elderly and MCI are statistically indistinguishable. Healthy aging appears to capture most of the effect; MCI does not add a further detectable decrement.
- No MoCA correlation. Within Elderly+MCI ($N = 32$), no ISFS scalar tracks MoCA (all $|r| \leq 0.13$).

References

1. **Lecci S**, Fernandez LMJ, Weber FD, Cardis R, Chatton JY, Born J, Lüthi A. (2017). Coordinated infraslow neural and cardiac oscillations mark fragility and offline periods in mammalian sleep. *Science Advances* 3(2): e1602026.
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2. **Lázár ZI**, Dijk DJ, Lázár AS. (2019). Infraslow oscillations in human sleep spindle activity. *Journal of Neuroscience Methods* 316: 22–34. <https://doi.org/10.1016/j.jneumeth.2018.12.002>
3. **Osorio-Forero A**, Cardis R, Vantomme G, Guillaume-Gentil A, Katsioudi G, Devenoges C, Fernandez LMJ, Lüthi A. (2021). Noradrenergic circuit control of non-REM sleep substates. *Current Biology* 31(22): 5009–5023.e7.
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4. **Dimitriades ME**, Osorio-Forero A, Fattinger S, von Arx S, Kurth S, Pugin F, Jaramillo V, Volk C, Krugliakova E, Furrer M, Leach S, Achermann P, Gerstenberg M, Huber R. (2024). The infraslow fluctuation of sigma power during sleep: links to markers of arousal and memory reactivation across development. *bioRxiv* 2024.11.06.620875. <https://doi.org/10.1101/2024.11.06.620875>